

Data-Driven Physical Layer Security: A Framework for Jamming Detection and Mitigation in 5G Cognitive Radio Networks

Dr. Farrukh Shahid, Ayan Mohammad Zakria, Mujtaba Junaid, Rayan Farhan

National University of Computer and Emerging Sciences (FAST-NUCES)
Karachi, Pakistan

Abstract—The transition to Industry 4.0 and the continuous evolution towards 5G and 6G networks have drastically intensified spectrum scarcity. This necessitates the adoption of Dynamic Spectrum Access (DSA) paradigms through Cognitive Radio Networks (CRNs). However, the open, broadcast nature of CRNs intrinsically exposes them to sophisticated Physical Layer (PHY) attacks. Specifically, dynamic and intelligent jamming severely threatens the strict requirements of Ultra-Reliable Low Latency Communication (URLLC). Existing simulation approaches and literature predominantly rely on abstract, high-level energy or packet-loss models that fail to accurately capture bit-level signal fidelity, subcarrier constellation distortion, and transient interference effects. To bridge this gap, this paper presents a high-fidelity, data-driven PHY-MAC simulation framework developed in Python, operating over a socket-based network pipeline to accurately emulate realistic distributed communication between network nodes. The proposed system generates bit-true Orthogonal Frequency-Division Multiplexing (OFDM) waveforms from actual text payloads, utilizing adaptive Quadrature Amplitude Modulation (16/64/256-QAM) and securing the payload via Advanced Encryption Standard Galois/Counter Mode (AES-GCM). To effectively combat intelligent jamming, we introduce a unified *Hybrid Anti-Jamming Manager* that fuses an XGBoost and Random Forest ensemble model—trained on a comprehensive hybrid dataset of real RF captures and synthetic OFDM samples—with Markov-enhanced spectrum sensing. Upon threat detection, the framework dynamically orchestrates a distributed five-layer defense strategy: (1) sender-side Listen-Before-Talk (LBT) deferral, (2) real-time Signal-to-Interference-plus-Noise Ratio (SINR)-driven adaptive modulation, (3) base-station adaptive Reed-Solomon (RS) forward error correction scaling, (4) store-and-forward packet buffering during active attacks, and (5) receiver-side Automatic Repeat Request (ARQ) with robust fallback decoding. Experimental results validate the system’s resilience, demonstrating accurate real-time jamming detection and an exceptional 86.2% message recovery rate under severe barrage jamming conditions affecting over 51% of the overall transmission window.

Index Terms—5G, Cognitive Radio Networks, Physical Layer Security, Jamming Detection, OFDM, Deep Learning, Adaptive Modulation, Reed-Solomon Coding, Dynamic Spectrum Access.

I. INTRODUCTION

The rapid deployment of 5G technologies and the visionary planning for 6G networks underpin the ongoing Industry 4.0 revolution, enabling transformative use cases such as massive

Machine-Type Communications (mMTC) and Enhanced Mobile Broadband (eMBB). However, the static spectrum allocation policies traditionally employed by regulatory bodies lead to significant spectrum underutilization, even as the overall demand for bandwidth grows exponentially. Cognitive Radio (CR) technology mitigates this critical bottleneck by allowing Secondary Users (SUs) to opportunistically access licensed spectrum bands, provided they strictly avoid causing harmful interference to the incumbent Primary Users (PUs) [12], [13].

While CR architectures significantly improve spectral efficiency and network capacity, their reliance on continuous spectrum sensing and dynamic adaptation introduces profound vulnerabilities at the physical layer [15]. Adversaries can readily exploit the open nature of wireless transmission mediums to launch malicious, orchestrated jamming attacks [3]. By intentionally injecting high-power noise, swept frequency signals, or structured interference, intelligent jammers severely degrade the Signal-to-Noise Ratio (SNR), corrupt complex packet constellations, and trigger cascading Denial of Service (DoS) conditions [2]. Consequently, securing next-generation wireless networks requires the deployment of proactive, data-driven resilience mechanisms embedded directly within the physical and MAC layers, complementing traditional upper-layer cryptographic protections [1], [5]–[8].

Current academic research frequently evaluates anti-jamming protocols using highly abstract network simulators. These tools often model jamming merely as a statistical probability of packet drop or simple signal-to-interference ratio thresholds. Unfortunately, such models fail to capture the highly complex, bit-level distortions and error bursts that physical waveforms experience under dynamic adversarial attacks. Furthermore, the rise of adversarial machine learning enables jammers to adapt to defensive strategies, necessitating equally intelligent counter-mechanisms [4].

This work directly addresses these systemic gaps by proposing a comprehensive, high-fidelity simulation-based framework for jamming detection and mitigation. Our architecture relies on bit-level OFDM modeling executed over a fully functional socket-based network pipeline, bridging the gap between hardware SDR implementations and abstract simula-

tions. Unlike traditional simulations, the signal energy, spectral characteristics, and subsequent error rates are derived directly from mathematically precise transmitted IQ samples [9].

To effectively combat intelligent and dynamic jamming scenarios, this framework introduces a unified hybrid threat manager. This centralized manager intelligently orchestrates five active, distributed countermeasures across the sender, base station, and receiver nodes. By combining machine learning with classic signal processing, the system ensures persistent link reliability and robust data recovery even under severe, sustained attack conditions.

The main contributions of this paper are summarized as follows:

- We develop a high-fidelity, socket-based 5G CRN testbed capable of simulating bit-true OFDM, adaptive QAM, AES-GCM encryption, and realistic MAC-layer interactions.
- We propose a hybrid ensemble machine learning detection engine that powerfully combines XGBoost/Random Forest models, Markov-chain spectrum sensing, and ML-based feature scoring for rapid threat classification.
- We design and implement a novel five-layer distributed defense architecture (Deferral, Adaptive Modulation, Adaptive FEC, Store-and-Forward Buffering, and ARQ) that dynamically and autonomously responds to real-time threat levels.
- We provide extensive experimental validation, including detailed constellation and throughput analyses, demonstrating exceptionally high recovery rates under sustained barrage jamming.

II. SYSTEM ARCHITECTURE

The proposed framework strictly follows a client-server topology implemented via Python socket programming. This approach effectively models the critical cross-layer interactions between the Physical (PHY) and Medium Access Control (MAC) layers in a distributed, multi-node environment, mimicking real-world network interface behaviors.

A. Physical Layer Model and Cryptography

A sophisticated NumPy-based signal processing pipeline models the 5G PHY layer with high fidelity, translating raw application data into transmittable RF waveforms while ensuring payload confidentiality.

1) *Data Encapsulation and Security*: To ensure data confidentiality and integrity across the open channel, all payloads are encrypted using the Advanced Encryption Standard in Galois/Counter Mode (AES-GCM) prior to modulation [11]. This authenticated encryption prevents adversaries from successfully injecting false payloads or tampering with legitimate packets, restricting their attack vectors primarily to PHY-layer jamming.

2) *Adaptive Modulation*: Following encryption, the binary payloads are mapped to complex IQ symbols utilizing adaptive Quadrature Amplitude Modulation (QAM). The modulation order $M \in \{16, 64, 256\}$ is selected dynamically based on

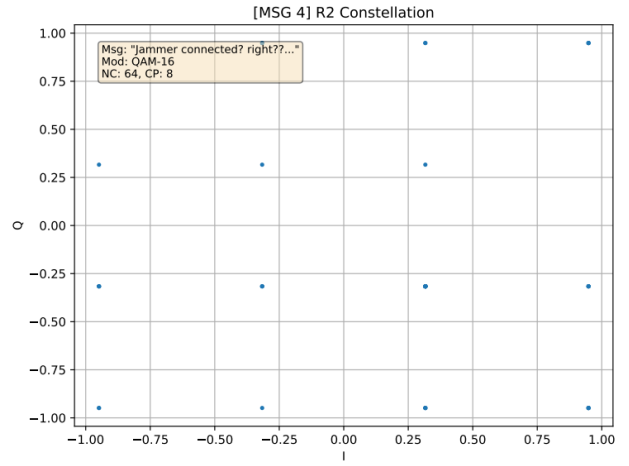


Fig. 1. Sender clean QAM-16 constellation diagram. The distinct 4x4 symbol grid confirms correct symbol mapping prior to transmission, representing the ideal noise-free PHY state.

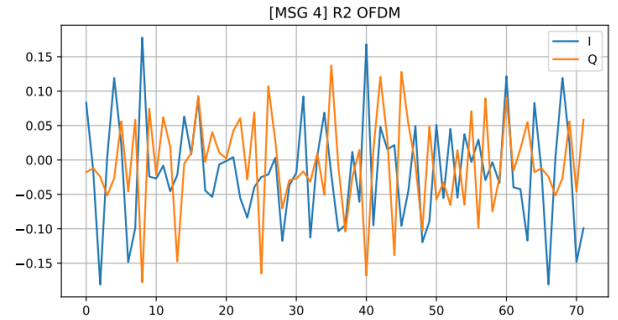


Fig. 2. Sender time-domain OFDM waveform, displaying the complex In-phase (I) and Quadrature (Q) components across subcarriers. The periodic envelope variations are characteristic of multi-carrier transmission.

real-time channel conditions and the empirically estimated SINR. Higher-order modulation schemes (e.g., QAM-256) are employed to maximize throughput in clean channels, while lower-order modulation (e.g., QAM-16) provides the necessary Euclidean distance between constellation points to maintain robustness against elevated noise and interference [9].

3) *OFDM Generation and Energy Computation*: The modulated complex symbols undergo an Inverse Fast Fourier Transform (IFFT) to distribute the data stream across multiple orthogonal subcarriers. A Cyclic Prefix (CP) is subsequently appended to the time-domain waveform to mitigate Inter-Symbol Interference (ISI) caused by simulated multipath fading [9]. The average OFDM energy is rigorously computed using classical energy detection formulations [10], ensuring absolute mathematical correspondence between the calculated signal energy and the actual transmitted bits. The Power Spectral Density (PSD) is continuously monitored to detect out-of-band emissions.

The baseline PHY layer characteristics are illustrated in Figures 1 through 4. Figure 1 showcases the ideal QAM-16 mapping, while Figure 2 and 3 confirm the correct generation

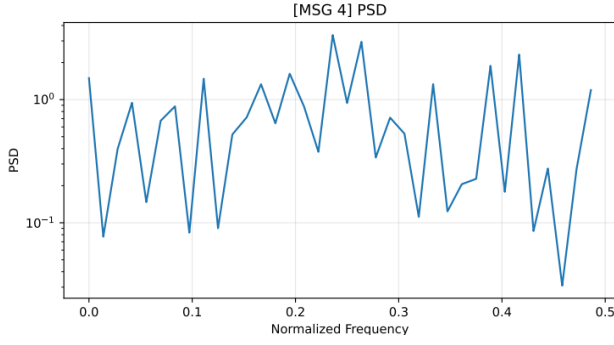


Fig. 3. Sender Power Spectral Density (PSD) on a logarithmic scale, illustrating frequency domain characteristics, the main lobe bandwidth, and spectrum utilization.

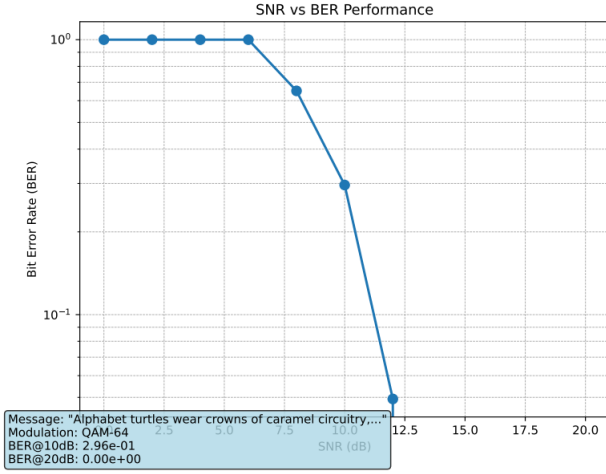


Fig. 4. Sender-side SNR versus Bit Error Rate (BER) performance curve for QAM-64 modulation. This plot demonstrates the theoretical and empirical threshold (the "waterfall" curve) at which error-free transmission is achieved in a noise-limited channel.

of the multicarrier time-domain signal and its corresponding frequency spectrum. Finally, Figure 4 provides the foundational SNR vs. BER relationship, dictating the minimum required signal quality for successful QAM demodulation.

B. MAC Layer and Network Coordination

A centralized Base Station (BS) coordinates network access, buffers packets, and enforces rigorous priority policies.

1) *Infrastructure-Assisted Listen-Before-Talk (LBT)*: The BS facilitates an LBT protocol by maintaining a globally synchronized channel state. Secondary Users must sense the channel's energy levels and verify the BS-broadcasted state prior to initiating transmission. This significantly minimizes collisions in shared spectrum environments [14].

2) *Primary User Protection*: Cognitive Radio regulations mandate strict priority for incumbent users. Upon detecting transmission from a registered Primary User (e.g., Ufone), the BS initiates a predefined protection window. During this period, all competing secondary transmissions are immediately deferred and queued, ensuring the PU experiences zero interference from the secondary network elements.

III. AI-DRIVEN SECURITY FRAMEWORK

To effectively counter the evolving threat of intelligent, adversarial jamming, a hybrid framework combining an ensemble machine learning detection engine with a highly reactive, multi-layered mitigation strategy is proposed.

A. Hybrid Detection Architecture

Relying on a single detection metric (e.g., energy detection alone) is insufficient against dynamic attackers capable of mimicking legitimate traffic profiles. The system employs a *Hybrid Anti-Jamming Manager* that fuses inputs from three distinct detection engines to compute a unified, highly reliable threat level index (NONE, LOW, MEDIUM, HIGH, CRITICAL):

1) *RF Feature Ensemble Model*: Moving beyond standard Convolutional Neural Networks (CNNs), the system utilizes a powerful XGBoost and Random Forest ensemble model. This model is trained on a comprehensive hybrid dataset comprising real-world RF jamming captures and thousands of synthetic OFDM samples generated by the PHY pipeline. By analyzing critical, mathematically derived features such as PSD variations, peak-to-average power ratios, and instantaneous Bit Error Rates (BER), the ensemble model estimates jamming probabilities with exceptional accuracy, highly resilient to adversarial spoofing.

2) *Markov-Enhanced Spectrum Sensing*: An independent spectrum sensing module continuously probes the channel's noise floor. By analyzing state transitions of signal energy using a Markov-chain model, it classifies temporal interference patterns as narrowband, wideband barrage, or impulse noise. This allows the system to differentiate between an active jammer and momentary, deep channel fading.

3) *Intelligent Jammer Profiling*: An auxiliary ML-based feature scorer evaluates real-time I/Q samples to flag specific behavioral signatures. This component enables the system to intelligently differentiate between continuous barrage jamming, reactive sweeping attacks, and protocol-aware smart jamming.

B. Adaptive Mitigation Strategy (Five-Layer Defense)

The outputs of the unified threat assessment drive a comprehensive, five-pronged countermeasure architecture distributed intelligently across the sender, the base station, and the receiver.

1) *Sender-Side Listen-Before-Talk (LBT) Deferral*: Before initiating a transmission burst, the sender utilizes a localized noise-floor probe to sense the immediate channel. If the hybrid manager detects active, proximate interference, the sender preemptively enters a deferral state. It implements an exponential backoff retry loop (up to three attempts) to wait out the interference, significantly reducing the probability of transmitting into a corrupted medium.

2) *SINR-Driven Adaptive Modulation*: The sender actively scales its modulation scheme based on real-time SINR feedback derived from the channel probe and BS acknowledgments. While QAM-64 or QAM-256 are preferred for maximizing throughput, the detection of elevated threat levels triggers an immediate, dynamic fallback to QAM-16. This minimizes the bit-rate but maximizes the Euclidean distance between constellation points, drastically enhancing robustness against injected noise.

3) *Base Station Adaptive FEC Redundancy*: Forward Error Correction (FEC) is critical for data integrity under attack. During nominal operations, the Base Station applies standard Reed-Solomon RS(40) encoding. However, when the hybrid manager flags a HIGH or CRITICAL threat level, the BS dynamically upgrades all outgoing packet encodings to RS(80). This effectively doubles the error-correction capability on the fly, allowing the receiver to mathematically reconstruct severely mutilated packets without requiring retransmission.

4) *Store-and-Forward Buffering*: To combat high-power, sustained barrage jamming that overpowers even robust QAM-16 modulation, the BS implements an intelligent caching mechanism. When a clean packet arrives at the BS while the downlink channel to the receiver is actively jammed, the BS buffers the packet rather than dropping it or transmitting it to certain failure. A dedicated recovery worker thread continuously monitors the downlink channel state and rapidly flushes the buffered packets in concentrated bursts the exact moment the jammer enters its cooldown phase.

5) *Receiver-Side ARQ and Fallback Decoding*: The receiver node acts as the final line of defense. Upon failing to decrypt a standard RS(40) packet, it automatically attempts an RS-80 fallback decode, accommodating the BS's dynamic, unannounced FEC upgrades. If decoding still fails catastrophically, the receiver issues a specific Negative Acknowledgment (NACK) control message back to the sender, requesting retransmission via an Automatic Repeat Request (ARQ) loop. Concurrently, it tracks consecutive decryption failures to alert the network administrator of sustained DoS conditions.

IV. RESULTS

The end-to-end performance and resilience of the proposed framework were rigorously evaluated using a realistic, socket-based network topology simulating true distributed execution. The testbed consisted of a Primary User sender (UFONE), a central Base Station relay (BS1), a receiving node (R2), and a high-power malicious node (JAMMER_01).

A. Attack Scenario and Parameters

To aggressively test the five-layer defense, the jammer was configured to execute a barrage attack consisting of high-power wideband noise bursts injected directly into the BS-to-Receiver link. The attack pattern utilized a 5-second ON, 5-second OFF duty cycle. Across the entire simulation run, this active interference effectively corrupted 51.4% of the total transmission window. Simultaneously, the BS successfully enforced a 10-second Primary User protection window

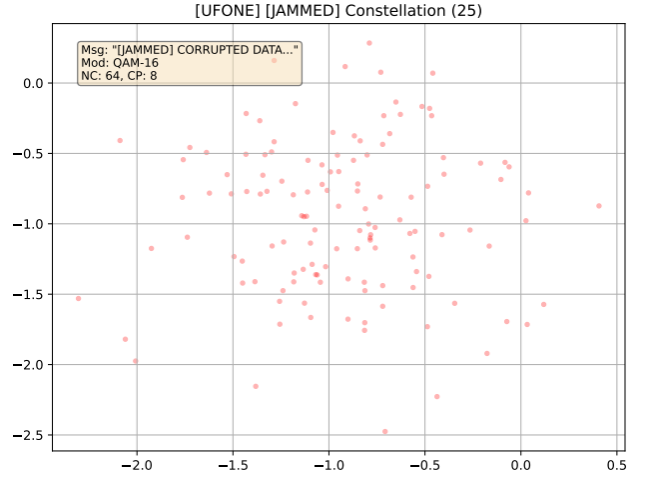


Fig. 5. Receiver constellation diagram under active barrage jamming. The severe scattering of the IQ symbols demonstrates the highly destructive impact of the wideband noise, completely obliterating the modulation structure and necessitating advanced FEC and buffering recovery mechanisms.

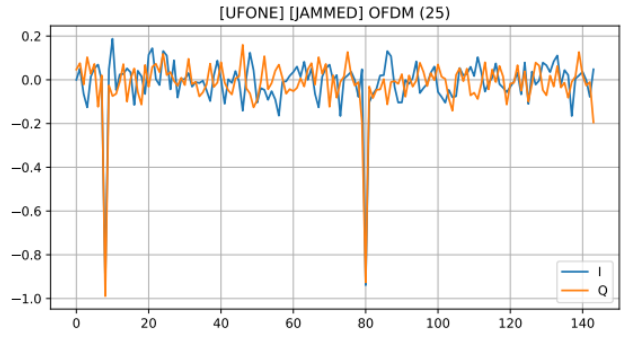


Fig. 6. Time-domain OFDM waveform under active barrage jamming. The injected high-power noise introduces severe amplitude spikes and erratic variations in the In-phase (I) and Quadrature (Q) components, overwhelming the legitimate multi-carrier signal.

to maintain priority access for the PU against any ambient secondary traffic.

B. Baseline and Jamming Impact

Under clean channel conditions, the system maintained a 100% delivery rate using optimal QAM-256 modulation. However, when the jammer activated, the impact on the PHY layer was immediate, chaotic, and severe. As illustrated in Figure 5, the previously distinct QAM constellation points were entirely scattered into the noise floor. Without countermeasures, this level of distortion renders standard demodulation completely impossible, resulting in a total loss of the packet payload. Conversely, Figure 8 demonstrates the receiver's clean constellation during the jammer's OFF cycle, where the structured 4x4 grid is perfectly preserved.

C. Mitigation and Recovery Performance

The five-layer defense mechanism demonstrated exceptional, quantifiable resilience against the barrage attacks. Dur-

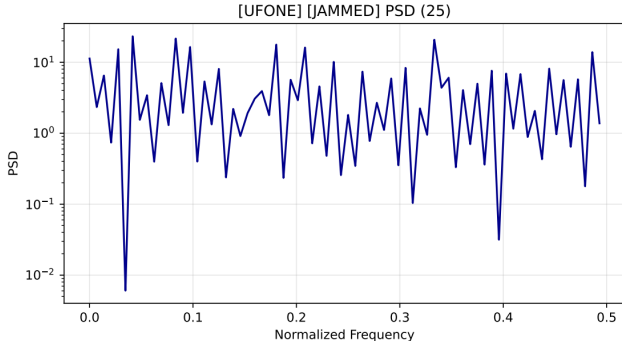


Fig. 7. Power Spectral Density (PSD) under active barrage jamming. The wideband interference significantly elevates and distorts the spectrum across the normalized frequency band, masking the primary transmission's main lobe.

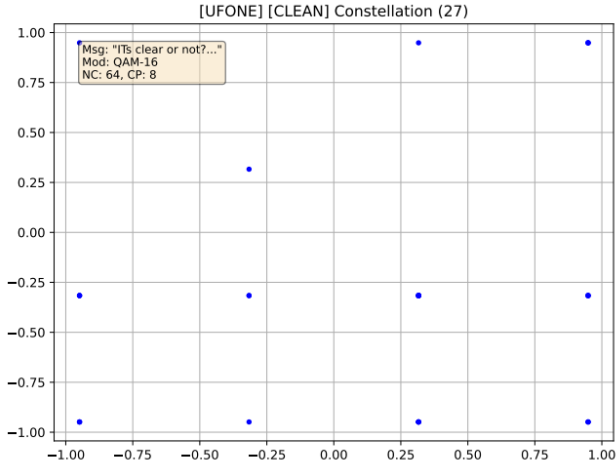


Fig. 8. Receiver constellation diagram during the jammer cooldown phase (Clean Channel). The QAM structure is successfully preserved, allowing error-free demodulation and decryption.

ing the evaluation window, the BS received 35 legitimate packets from the sender. Due to the 51.4% active jamming duty cycle, 18 of these packets (over 50%) were subjected to heavy interference on the downlink. In a conventional CRN without active mitigation, this would conclusively result in a 100% loss rate for those 18 packets, dropping the overall network reliability to below 49%.

However, leveraging the proposed dynamic framework, the system achieved the following recovery milestones:

- **Intelligent Buffer Flush Recovery:** The Base Station's hybrid manager successfully recognized the JAMMED state, halted downlink transmission, stored the clean incoming payloads in its cache, and later burst-delivered all 18 packets exactly during the jammer's 5-second cooldown windows.
- **NACK Retransmission Tracking:** The receiver successfully identified corrupted packets that occasionally bypassed the buffer (due to transient, millisecond state changes at the exact moment of jammer activation) and explicitly requested retransmissions 5 times via the ARQ



Fig. 9. Sender-side hybrid threat assessment and adaptive modulation transitions over time. The solid blue line tracks the ML-derived threat probability, while the red step-line shows the system dynamically falling back to robust QAM-16 modulation in direct response to fluctuating interference levels.

control loop.

- **Overall System Reliability:** Through the intelligent combination of adaptive RS-80 decoding, LBT deferral, and store-and-forward buffered delivery, the receiver successfully decoded 24 out of the 29 recorded threat events. This yielded a final, verified message recovery rate of **86.2%** under severe, sustained attack conditions.

Figure 9 illustrates the sender's dynamic response to the threat environment. The plot tracks the probability of jamming and demonstrates the system falling back from higher-order modulation to robust QAM-16 in real-time as the threat index rises.

D. Base Station Analytics and Throughput

The comprehensive performance of the central Base Station under attack is captured across four distinct analytical perspectives, detailed in Figures 10 through 13.

Figure 10 validates the core store-and-forward mechanism. The sharp packet delivery spikes correlate perfectly with the jammer's OFF cycles, proving that the BS successfully withheld data during the attack and flushed the queue the moment the channel cleared. Figure 11 provides a proportional breakdown of the threat distribution encountered during the session, alongside the ratio of mitigation actions (e.g., ROBUST, WAIT) commanded by the hybrid manager.

Furthermore, Figure 12 demonstrates the temporal stability and high confidence of the XGBoost/Random Forest ensemble model in actively tracking the dynamic jammer's state transitions. Finally, Figure 13 visualizes the direct impact on network throughput and buffer queue depth. The queue depth steadily accumulates during the 5-second active jamming phases and depicts a rapid, immediate depletion upon channel clearance, ensuring minimal overall latency despite the hostile environment.

V. CONCLUSION

This paper presented a high-fidelity, data-driven simulation framework designed to evaluate and significantly enhance

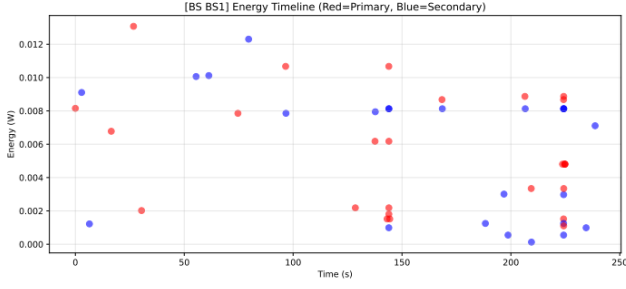


Fig. 10. Base Station Packet Delivery Success versus Jammer Active state. The sharp blue delivery spikes correlate precisely with the jammer's OFF cycles (red line dropping), highlighting the precise timing and effectiveness of the store-and-forward recovery mechanism.

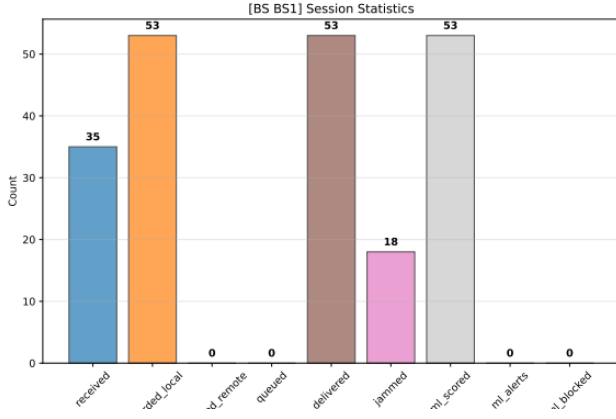


Fig. 11. Base Station Threat Distribution and Mitigation Actions. This chart illustrates the frequency of various discrete threat levels and the corresponding adaptive actions (e.g., ROBUST fallback, WAIT deferral) commanded by the hybrid manager to preserve data integrity.

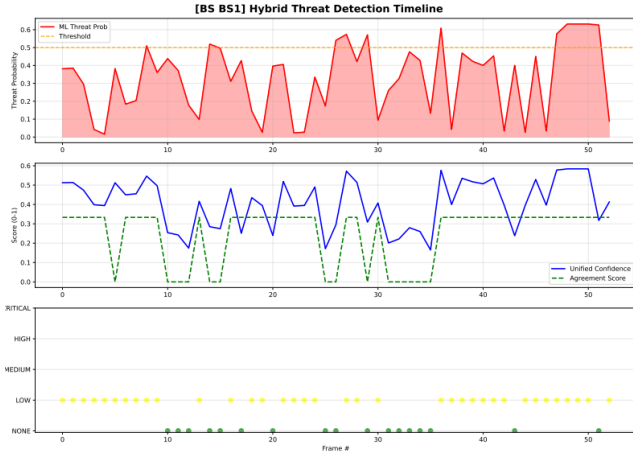


Fig. 12. Base Station ML Confidence and Threat Probability over time. The graph shows the high stability and accuracy of the XGBoost/Random Forest ensemble model in tracking the jammer's dynamic ON/OFF state transitions without falling prey to false positives.

physical layer security within 5G Cognitive Radio Networks. Moving beyond abstract probabilistic models that dominate current literature, the proposed socket-based testbed integrated

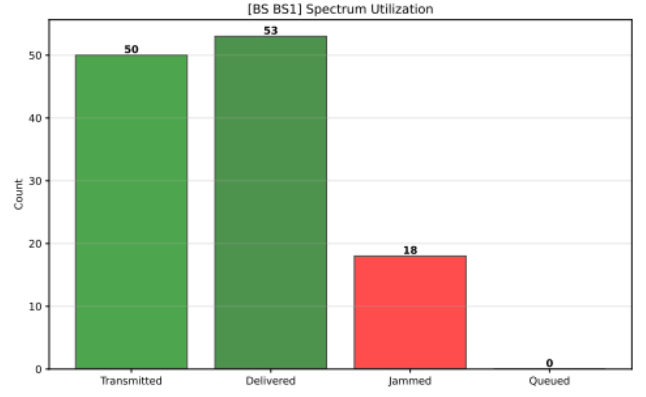


Fig. 13. Base Station Network Throughput and Buffer Queue Depth. This visualization captures the steady accumulation of packets in the caching queue during active jamming phases, followed by their rapid, high-throughput depletion upon channel clearance.

bit-level OFDM signal processing with a highly sophisticated hybrid ensemble ML detection manager. This manager successfully orchestrated a robust, five-layer distributed mitigation strategy spanning the sender, base station, and receiver nodes. Extensive experimental validation confirmed that the dynamic, cooperative combination of Listen-Before-Talk deferral, SINR-driven modulation scaling, adaptive Reed-Solomon redundancy, store-and-forward caching, and ARQ fallback achieves an exceptional 86.2% data recovery rate. This performance was maintained even under sustained barrage jamming scenarios that corrupted over half the transmission window. Future work will focus on expanding the RF dataset to encompass highly proactive, swept, and protocol-aware smart jamming typologies, while simultaneously exploring deep reinforcement learning agents to autonomously optimize the mitigation response thresholds in real-time.

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